

Week 3 Task 2 Report: User Account Management in Linux

Objective

The purpose of this task is to create and manage user accounts on a Linux system by assigning appropriate permissions based on user roles. This is essential for enforcing the principle of least privilege and maintaining a secure operating environment.

Accounts Created

Two users were created for demonstration purposes:

Username	Role	Privileges
analyst	Admin-level	Sudo access (can run admin tasks)
intern	Limited	Standard user with no sudo rights

Step-by-Step Implementation

1. Create New User Accounts

```
sudo adduser analyst
```

```
sudo adduser intern
```

These commands create user accounts and prompt for password setup.

2. Assign Role-Based Permissions

► Grant analyst user administrative rights:

```
sudo usermod -aG sudo analyst
```

This adds the analyst user to the sudo group, enabling administrative access.

► Confirm group memberships:

```
groups analyst
```

```
groups intern
```

These commands verify that the users are assigned to their appropriate groups.

✓ 3. Create a Custom Group for Intern

To demonstrate further control, a custom group limited was created:

```
sudo groupadd limited
```

```
sudo usermod -aG limited intern
```

This enables finer-grained access management in future configurations.

✓ 4. Directory-Level Access Control

A secure directory was created for the analyst user to demonstrate permission handling:

```
sudo mkdir /opt/securedata
```

```
sudo chown analyst:analyst /opt/securedata
```

```
sudo chmod 700 /opt/securedata
```

This restricts access to the directory such that only analyst can view or modify its contents.

✓ 5. Testing User Access and Privileges

Switch to each user to verify role-based access:

```
su - analyst
```

```
sudo apt update (work with password prompt)
```

```
su - intern
```

```
sudo apt update (fail cuz user is not in the sudoers file)
```

These tests validated correct user role assignment and access limitations.

✓ Outcome

- User accounts were successfully created and configured with appropriate permissions.
- Administrative and limited-access roles were clearly enforced.
- File system permissions were tested and verified using a secure directory.
- All activities were logged and executed as per best practices in Linux system administration.